

Quantum Computation and Information

Homework Set 1

1. Given a state of a qubit $|\psi\rangle = a|0\rangle + b|1\rangle$, where $a, b \in \mathbb{C}$ and $|a|^2 + |b|^2 = 1$, and recall that states $|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$.

(a). In 2-dimensional complex space with basis $\{|0\rangle, |1\rangle\}$, write down the matrix form of operator $A = |+\rangle\langle+| + \frac{1}{2}|-\rangle\langle-|$, and verify A is an observable (i.e., self-adjoint).

Solutions: With basis $\{|0\rangle, |1\rangle\}$, $|+\rangle = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix}$ and $|-\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$. Thus,

$$A = \begin{pmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}.$$

Note that $A = A^\dagger$, so it is self-adjoint.

(b). If you measure the observable A on the state $|\psi\rangle$, what are the possible measurement outcomes and the associated probabilities?

Solutions: Note that observable A has eigenvectors $|+\rangle$ and $|-\rangle$ with eigenvalues 1 and 1/2, respectively. So, the measurement outcome will be 1 with probability $|\langle+|\psi\rangle|^2 = |\frac{a}{\sqrt{2}} + \frac{b}{\sqrt{2}}|^2 = \frac{1}{2}|a+b|^2$, and $\frac{1}{2}$ with probability $|\langle1|\psi\rangle|^2 = |\frac{a}{\sqrt{2}} - \frac{b}{\sqrt{2}}|^2 = \frac{1}{2}|a-b|^2$.

2. Consider a state of two qubits $|\psi\rangle = \frac{1}{2}|00\rangle + \frac{1}{2}|01\rangle - \frac{i}{\sqrt{2}}|11\rangle$.

(a). If you label the basis of 4-dimensional complex space in the following order: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, what are the vector forms of $|\psi\rangle$ and $\langle\psi|$?

Solutions: $|\psi\rangle = \begin{pmatrix} 1/2 \\ 1/2 \\ 0 \\ -i/\sqrt{2} \end{pmatrix}$ and $\langle\psi| = (1/2 \quad 1/2 \quad 0 \quad i/\sqrt{2})$.

(b). If you label the basis of 4-dimensional complex space in the following order: $|00\rangle, |10\rangle, |01\rangle, |11\rangle$, what are the vector forms of $|\psi\rangle$ and $\langle\psi|$?

Solutions: $|\psi\rangle = \begin{pmatrix} 1/2 \\ 0 \\ 1/2 \\ -i/\sqrt{2} \end{pmatrix}$ and $\langle\psi| = (1/2 \quad 0 \quad 1/2 \quad i/\sqrt{2})$.

(c) If you label the basis of 4-dimensional complex space in the following order: $|00\rangle, |01\rangle, |10\rangle, |11\rangle$, what is the matrix form of density operator $\rho = |\psi\rangle\langle\psi|$?

Solutions: $\rho = \begin{pmatrix} 1/2 & 1/2 & 0 & i/\sqrt{2} \\ 1/2 & 1/2 & 0 & i/\sqrt{2} \\ 0 & 0 & 0 & 0 \\ -i/\sqrt{2} & -i/\sqrt{2} & 0 & 1/2 \end{pmatrix} (1/2 \ 1/2 \ 0 \ i/\sqrt{2}) = \begin{pmatrix} 1/4 & 1/4 & 0 & i/2\sqrt{2} \\ 1/4 & 1/4 & 0 & i/2\sqrt{2} \\ 0 & 0 & 0 & 0 \\ -i/2\sqrt{2} & -i/2\sqrt{2} & 0 & 1/2 \end{pmatrix}.$

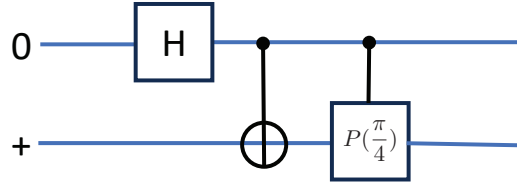
(d) Is $|\psi\rangle$ a separable state or entangled state?

Solutions: It is an entangled state. Suppose otherwise, then there exist a, b, c, d such that

$$|\psi\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle,$$

and thus $ac = 1/2$, $ad = 1/2$, $bc = 0$, $bd = -1/\sqrt{2}$, which however has no solution.

3. Recall the quantum gate notation and diagrams introduced in lecture 2, and consider the following circuit with the input $|0\rangle|+\rangle$.



(a). What is the output of the above circuit?

Solutions: $|0\rangle|+\rangle \Rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|+\rangle \Rightarrow \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)|+\rangle \Rightarrow \frac{1}{\sqrt{2}}|0\rangle|+\rangle + \frac{1}{2}|1\rangle(|0\rangle + e^{i\pi/4}|1\rangle) = \frac{1}{\sqrt{2}}|+\rangle|0\rangle + \frac{1}{2}(|0\rangle + e^{i\pi/4}|1\rangle)|1\rangle.$

(b). We measure the second qubit (denoted by the lower horizontal line in the circuit) on the computational basis $\{|0\rangle, |1\rangle\}$ at the end of computation. After the measurement, what are the possible states of the first qubit, and the associated probabilities?

Solutions: The first qubit will be at state $|+\rangle$ if the second qubit is at state $|0\rangle$ after measurement, and $\frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}|1\rangle)$ if the second qubit is at state $|1\rangle$ after measurement, with equal probability.